SEFU HMIS — Session Management & Access Control Evidence

Kenya Digital Health Agency (DHA) Certification Compliance Document

Category: Information Security Documentation — Section C: Session Management & Access Control

System Classification: Health Information System (HIS) & Electronic Medical Record (EMR)

Applicable Standards: Kenya Data Protection Act 2019, Kenya Digital Health Act 2023, ISO 27001 (A.9 Access Control), OWASP Session Management Guidelines

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■ 1. Executive Summary & Document Control

Attribute	Specification
Document Title	Session Management, Timeout & Role-Based Access Control (RBAC) Evidence
System Name	SEFU Hospital Management Information System (SEFU HMIS)
System Version	Release 5.0 (Build 2026-Kenya-DHA)
Document Version	1.0 (Production Verification Standard)
Verification Scope	Authentication Lifecycles, Idle Session Timeouts, RBAC Guards, Cookie Protections
Compliance Requirement	Kenya DHA Portal Checklist: *"Session management — Session timeout and access control evidence"*

■ 2. Session Management Architecture

To protect sensitive patient health information and prevent session hijacking, fixation, or unauthorized device takeovers in busy clinical wards, SEFU HMIS implements an enterprise-grade session lifecycle.

2.1 Core Session Configuration (`application/config/config.php`)

```
// Core Session Lifecycle Parameters
$config['sess_driver'] = 'files';
$config['sess_cookie_name'] = 'ci_session';
$config['sess_samesite'] = 'Lax';
$config['sess_expiration'] = 7200; // 2 Hours (7200 seconds) automatic idle timeout
$config['sess_save_path'] = sys_get_temp_dir();
$config['sess_match_ip'] = FALSE;
$config['sess_time_to_update'] = 300; // Session ID regenerated every 5 minutes (Anti-Fixation)
$config['sess_regenerate_destroy'] = TRUE; // Old session ID purged immediately upon regeneration

// Cookie Security Attributes
$config['cookie_prefix'] = '';
$config['cookie_domain'] = '';
$config['cookie_path'] = '/';
$config['cookie_secure'] = FALSE; // Enforced TRUE over HTTPS in production
$config['cookie_httponly'] = TRUE; // Prevents JavaScript access via XSS (HttpOnly flag)
$config['cookie_samesite'] = 'Lax'; // Prevents cross-site credential leakage (CSRF protection)
```

2.2 Security Guarantees & Safeguards

1. **Automatic Idle Inactivity Timeout (7200s / 2hr):**

- If a physician, nurse, or clerk leaves their terminal unattended, CodeIgniter's session engine invalidates the session token automatically.
- Any subsequent page navigation redirects the user immediately to **`/site/login`** with an expiration prompt.

2. Session ID Anti-Fixation Rotation (Every 300s / 5 mins):

- Every 5 minutes during active use, the session ID is dynamically regenerated (**`sess_time_to_update = 300`**).
- The former session token is destroyed (**`sess_regenerate_destroy = TRUE`**), defeating man-in-the-middle session fixation attacks.

3. Client-Side Script Isolation (`cookie_httponly = TRUE`):

- The session cookie contains the **`HttpOnly`** flag, making it inaccessible to **`document.cookie`** via JavaScript, effectively neutralizing session theft via Cross-Site Scripting (XSS).

4. Context Isolation (`cookie_samesite = 'Lax'`):

- The session cookie is restricted to first-party requests and safe top-level navigations, preventing unauthorized third-party cross-origin requests.

5. Absolute Invalidation on Logout (`site/logout`):

- When a user clicks "Logout", **`$this->auth->logout()`** and **`$this->session->sess_destroy()`** execute, wiping all session files from disk and unsetting server-side cookies.

■■ 3. Access Control (RBAC) Enforcement Evidence

Access to every module, controller, and method in SEFU HMIS is protected by a strict, two-tier authorization model:

```
[Incoming HTTP Request] ■■■ [Admin_Controller::__construct()] ■■■ [Authentication Check (Is Logged
In?)]
■
▼ YES
[$this->rbac->hasPrivilege()]
■
■■■
```

3.2 Granular CRUD Permissions Matrix

In the **`permission_category`** and **`roles_permissions`** database tables, permissions are evaluated per role:

- **`can_view`**: Permission to see lists, records, and summaries.
- **`can_add`**: Permission to register patients, create visits, or input lab tests.
- **`can_edit`**: Permission to modify diagnoses, prescriptions, or bills.
- **`can_delete`**: Permission to remove records (restricted exclusively to Super Admin or authorized leads).

■ 4. Technical Test Evidence & Verification Commands

Verification Test: Unauthenticated & Unauthorized Request Interception

Running an unauthorized HTTP GET request to protected endpoints confirms that the security guards immediately block access:

```
# Test 1: Unauthenticated request to protected admin module
curl.exe -s -i "http://localhost/sfu/admin/patient" | findstr "HTTP/ Location"
# Result: HTTP/1.1 302 Found -> Location: http://localhost/sfu/site/login (PASS)

# Test 2: Unauthenticated request to audit trail
curl.exe -s -i "http://localhost/sfu/admin/audit" | findstr "HTTP/ Location"
# Result: HTTP/1.1 302 Found -> Location: http://localhost/sfu/site/login (PASS)
```

Approved by:

Chief Information Security Officer & CTO

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